JANMITRA

Civic Issue Reporting & Management System

Nagpur Municipal Corporation | 4 Categories | 4 Wards | 4 Departments

Team ResolveIT

Team Work Distribution & Execution Plan

Who does what, when, and how the team works together

CITIZEN PORTAL	Report Issue | Track Status | Verify Resolution
BACKEND (Django)	REST API | Auth | Business Logic | AI Module
AI VALIDATION	Fake Detection (EXIF+ELA+HuggingFace) | YOLO Relevance Check
DATABASE	PostgreSQL + PostGIS | 4 Categories | 4 Wards | 4 Depts
DEPT. DASHBOARD	PWD | Solid Waste | Water Works | Electrical

Anas Shaikh | Anurag Deshmukh | Om Gadekar | Faizan Rayeen

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1. Work Division Overview

The project is divided into 4 responsibility areas, one per team member. Each area has both backend and frontend work. Parallel development is possible in most phases. Anas builds the foundation; others build on it.

ANAS SHAIKH -- Backend Core + Infrastructure Lead

- Django project setup: all 4 apps, settings (base/local/production), URL routing
- Custom User model with ward + department + designation + employee_id fields
- Registration and Login APIs with JWT token generation
- RBAC permission classes: IsCitizen, IsWorker, IsDeptOfficer, IsAdmin
- Category model (4 entries), Ward model (4 entries), Department model (4 entries)
- Complaint model with all fields including fake_detected, ward_override, override_reason
- Complaint CRUD APIs, status update API
- complaints/validators.py: validate_worker_assignment() function with override logic
- AssignWorkerView API with dept+ward validation + admin override support
- auto_close_unverified Celery periodic task (every 6 hours)
- Docker Compose: all 6 services configured and tested on all team machines
- Render deployment: backend web service + Celery background worker
- Database migration management and seed_demo management command
- Backend unit tests: auth, complaint CRUD, assignment validation
- Postman collection for all 24 API endpoints

ANURAG DESHMUKH -- AI Module + Dashboard Backend Lead

- complaints/ai_module/exif_checker.py -- Layer 1 fake detection step 1
- complaints/ai_module/ela_checker.py -- Layer 1 fake detection step 2
- complaints/ai_module/fake_detector.py -- HuggingFace AI fake detector step 3
- complaints/ai_module/yolo_validator.py -- Layer 2 YOLO relevance check
- complaints/ai_module/duplicate_detector.py -- PostGIS same-category+ward 100m check
- complaints/ai_module/priority_engine.py -- weighted 4-factor scoring formula
- complaints/tasks.py -- full AI pipeline Celery task chain
- route_complaint Celery task: category -> department auto-routing
- Dashboard app: KPIView, WardStatsView, TrendsView, WorkerWorkloadView, HighPriorityView
- PendingValidationView + manual approve/reject endpoint
- Dashboard filters: date range, ward, status, severity (applied to all dashboard APIs)
- Civic issue dataset download from Roboflow (pothole, garbage, water drain datasets)
- YOLO fine-tuning on Google Colab using civic dataset
- AI module unit tests: fake detection thresholds, duplicate radius, priority scoring
- Dashboard API tests: KPI counts, ward stats accuracy, dept isolation

OM GADEKAR -- Frontend Lead + Worker + Officer Portals

- React project: Vite + Tailwind CSS + React Router v6 setup
- AuthContext (React Context API): token, user, role, login, logout
- Axios client with JWT interceptor and auto-refresh on 401
- ProtectedRoute HOC: role-based route protection
- Design system: all shared components (StatusBadge, PriorityBadge, CategoryBadge, WardBadge, KPICard, DeptHeader)
- ImageUploader component with EXIF spinner and fake image error UI
- Worker TaskList: priority-sorted cards with dept+ward filtering (backend handles filter)
- Worker TaskDetail: Leaflet map with complaint pin, proof upload, Mark Done flow
- Officer Dashboard page: KPI cards, Ward-wise bar chart, Severity pie, Trend line, Worker workload table, High-priority

list
- Officer AssignWorker component: filtered worker dropdown + admin override UI
- Officer ValidationQueue: AI confidence bars, photo thumbnail, Validate/Reject buttons
- Admin CrossDeptDashboard: 4 dept comparison cards + cross-dept charts
- Admin WorkerManagement: list, register, edit workers (dept+ward selection)
- React deployment to Render Static Site
- Component tests for all shared components

FAIZAN RAYEEN -- Citizen Portal + Notifications Lead

- Login/Register page UI (all roles -- one page with role determined by backend)
- Citizen Home page: Report button, quick stats, recent complaints
- Complaint Submission Form: 4 category cards, ward selector (Ward 1-4), photo upload, Leaflet GPS map, submit
- My Complaints list with filter tabs (All/Open/In-Progress/Resolved)
- Complaint Detail with status timeline stepper (6 steps)
- Resolution Verification: proof photo gallery, Accept (star rating) / Reject (reason)
- Notification bell component: unread count badge, dropdown list, mark-all-read
- notifications/fcm_service.py: Firebase Admin SDK, send_push() function
- notifications/tasks.py: send_push_notification Celery task
- All FCM trigger points wired to status changes in Anas's complaint tasks
- firebase.js: Firebase init, FCM token request, foreground message handler
- firebase-messaging-sw.js: service worker for background push
- Demo data seed: citizens, workers (all 4 dept + ward combos), officers, complaints
- Cross-browser testing: Chrome, Firefox, Android Chrome
- README.md with setup guide, demo accounts table, API docs link

2. Timeline -- 40 Working Days

Phase	Days	Focus	Key Milestone
Phase 0	1-3	Setup	Docker runs on all machines. 4 categories, 4 wards, 4 depts seeded. API contract agreed.
Phase 1	4-10	Foundation	Auth works. Complaint submits with category+ward. AI layer 1 (EXIF) runs.
Phase 2	11-20	Core Workflow	Full AI pipeline. Dept+ward assignment. Worker completes complaint. Citizen verifies.
Phase 3	21-28	Dept. Dashboard	All 5 dashboard widgets live. Officer and admin portals complete.
Phase 4	29-36	Testing + Deployment	System live on Render. All roles tested. Bugs fixed. UptimeRobot active.
Phase 5	37-40	Demo Prep	Demo rehearsed. All 9 docs final. Seed data ready.

3. Git Workflow

Branch	Owner	Rule
main	Anas	Production-ready only. Merge from develop after team sign-off.
develop	All	Working branch. Feature branches merge here via Pull Request.
feature/anas/...	Anas	One branch per backend feature.
feature/anurag/...	Anurag	One branch per AI file or dashboard API.
feature/om/...	Om	One branch per React screen/component.
feature/faizan/...	Faizan	One branch per citizen screen or notification feature.

Pull Request rule: minimum 1 reviewer approval before merge to develop. Anas resolves all migration conflicts.

4. Critical Coordination Points

Day 3 (Phase 0)	All 4 agree on exact API contract: request/response format for every endpoint. Anas documents in Postman collection. All others build against this contract.
Day 7 (Phase 1)	Anas pushes: User model + Complaint model + Category/Ward/Dept models. All others can start building against these. Anurag needs Complaint model for AI module.
Day 12 (Phase 2)	Anurag pushes: full AI Celery pipeline. Faizan wires FCM triggers. Om can test assignment UI.
Day 22 (Phase 3)	Anurag pushes: all dashboard APIs. Om builds dashboard UI against these. Admin cross-dept view needs all dept APIs ready.
Day 30 (Phase 4)	Full integration test on Render. All 4 members test their own flows end-to-end on production URL.